An Examination of the Relationship between Financial Characteristics and Portfolio Performance Using Models from Backward Elimination and Principal Component Analysis

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Abstract

This report looks at traded company data obtained from Green Square Capital, and compares models using both backward elimination and principal component analysis to determine which 13 factors used to evaluate different companies are key indicators in predicting Portfolio Performance. The dataset from Green Square Capital consists of 33 traded companies across 8 sectors, along with 62 untraded companies. Price to Book Value, Earning per Share, and Free Cash Flow Margin are a few examples of the 13 factors that the management team uses to financially assess company performance. A full list of the 13 factors will be attached at the end of this summary report.

Keywords: Principal Component Analysis, PCA, Multiple Regression, Financial Modeling

1. Objective and Methods

The main goal of our analysis was to evaluate Portfolio Performance, and to create a model for Portfolio Performance that was in some way simpler than the 13 variable model used by Green Square Capital. We used two methods to achieve this goal: backward elimination and principal component analysis (PCA). In the former case, we would attempt to eliminate variables that were not statistically significant, thereby simplifying the model. With principal component analysis, we would keep all 13 predictor variables but aim for dimension reduction by using fewer than 13 principal components.

Additionally, we used Portfolio Performance as our response variable which we computed as "percent change in price per share between March 31 and June 30, 2016" found in the dataset from Green Square Capital with "beginning price at 3/31" and "ending price at 6/30".

2. Backward Elimination

The first method that we utilize is Backward Elimination, which is a statistical technique used in Regression Analysis. Under this approach, one starts with a model that contains all of the predictor variables of interest. Then, the least significant variable determined by the highest p-value is dropped, so long as it is not significant at our chosen critical level. We continue by successfully re-fitting reduced models and applying the same rule until all remaining variables are statistically significant or the desired adjusted R^2 is met.

The best model found by backward elimination used the following 5 predictor variables:

1. DPS 5 years (Dividend Per Share - 5 year growth rate)
2. DY (Dividend Yield)
3. TAT (Total Asset Turnover)
4. DEPS (Diluted Earnings per Share)

5. FCFM (Free Cash Flow Margin)

```
Call:
lm(formula = ds$Pct.Change ~ ds$DPS_5yrs + ds$DY + ds$TAT + ds$DEPS +
    ds$FCFM)

Residuals:
    Min      1Q  Median      3Q     Max
-14.932  -4.150   0.003   4.225  17.309

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   23.190      5.368   4.320 0.000189 ***
ds$DPS_5yrs    2.850      1.447   1.970 0.059221 .
ds$DY         -5.864      1.053  -5.571  6.6e-06 ***
ds$TAT       -25.552     14.767  -1.730 0.094996 .
ds$DEPS       -3.390      2.552  -1.329 0.195086
ds$FCFM       11.267      6.417   1.756 0.090454 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.768 on 27 degrees of freedom
Multiple R-squared:  0.6985,    Adjusted R-squared:  0.6427
F-statistic: 12.51 on 5 and 27 DF,  p-value: 2.425e-06
```

Figure 1: Best Regression Model on Traded Companies Using Backward Elimination

We set our alpha-criterion to be at the 0.05 level. It is important to note that when looking at the model for backward elimination, 3 predictors are statistically significant at the 0.1 alpha level and DEPS is not statistically significant. However, when we removed DEPS in the next iteration of the model, our adjusted R^2 began to drop. Since our goal was to maximize our ability to explain portfolio performance with the least number of predictor variables, it seemed prudent to keep these variables despite not being statistically significant. Therefore, we concluded that the multiple regression model using DPS_5years, DY, TAT, DEPS, and FCFM did the best job in predicting portfolio performance.

Furthermore, one limitation using regression analysis is that it is very sensitive to outliers. This is amplified by the relatively small dataset we were using (only 33 traded companies). When we looked at the residual plots for the model above, we saw that Archer Daniels Midland (ADM), was an outlier that had high leverage in our model. Since ADM had a relatively high influence on our multiple regression model, we removed this company and re-fit the model using the same backward elimination techniques. Looking at the below model, the adjusted R^2 increases drastically from 64% to 80%, and 2 out of the 5 predictors also changed. The significant predictors with the new model are:

1. DPS Today (Dividend Per Share - Current Quarter)
2. DY (Dividend Yield)

3. TAT (Total Asset Turnover)
4. PBV (Price to Book Value)
5. FCFM (Free Cash Flow Margin)

```
Call:
lm(formula = ds$Pct.Change ~ ds$PBV + ds$DPS_today + ds$DY +
    ds$TAT + ds$FCFM)

Residuals:
     Min       1Q   Median       3Q      Max
-11.4075  -1.8039  -0.2505   2.1661   9.4806

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   14.6796     4.5100   3.255 0.003143 **
ds$PBV         0.1516     0.1136   1.334 0.193684
ds$DPS_today   2.8734     0.8573   3.352 0.002467 **
ds$DY         -3.3580     0.7700  -4.361 0.000181 ***
ds$TAT       -61.9623     8.8657  -6.989 2.02e-07 ***
ds$FCFM       10.4869     4.6076   2.276 0.031321 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.924 on 26 degrees of freedom
Multiple R-squared:  0.8352,    Adjusted R-squared:  0.8035
F-statistic: 26.35 on 5 and 26 DF,  p-value: 2.091e-09
```

Figure 2: Best Regression Model on Traded Companies with ADM Removed Using Backward Elimination

Additionally, when the management team at Green Square Capital financially assesses a particular company, they calculate and assign an Index score to each company using the original 13 variables. They then eliminate any companies whose score is 5 or higher. As mentioned above, in the dataset that was provided to us we also received companies whose score was greater than 5, and we looked at the distributions of the traded companies vs untraded companies. The distribution of the untraded companies was much more normally fitted than the distribution of the traded companies. In addition, we expected that the standard deviation for the traded companies would be lower and the mean would be higher since Green Square Capital has historically been very successful at picking companies for their portfolios. However, we found exactly the opposite - the standard deviation of the traded companies was higher and the mean was lower. That being said, it is important to keep in mind that the scaling is not the same. There were 33 traded companies and 62 untraded companies. We were also only considering one quarter of data, and were not able to capture larger trends.

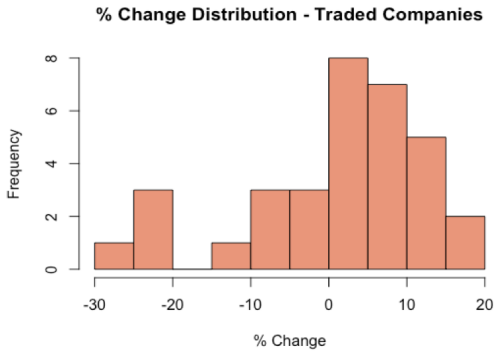


Figure 3: Distribution of Traded Companies

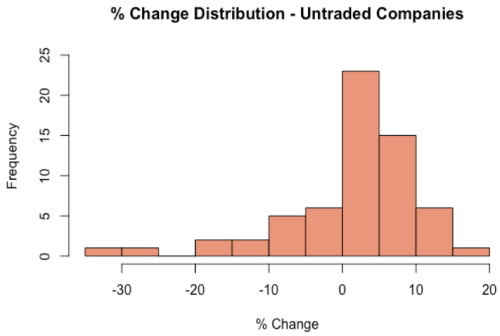


Figure 4: Distribution of Untraded Companies

3. Principal Component Analysis

Our second method was principal component analysis (PCA). Principal Component Analysis is a popular approach for deriving a low-dimensional set of features from a large set of variables. It is a technique for reducing the dimension of an n x p matrix X. The general method is calculate the covariance matrix of the original matrix X, and find the associated eigenvalues and eigenvectors. The eigenvectors will be orthonormal to each other. The largest eigenvalues and their corresponding eigenvectors are your principal components. The first principal component of the data is that along which the observations vary the most. In other words, the first principal component defines the line that is as close as possible to the data. It minimizes the sum of the squared perpendicular distances between each point and the line. In this way, the larger the eigenvalue of your principal component, the larger the amount of variability in the data is explained by that principal component. Principal components are ranked by the proportion of variability in the data they explain. As you add more principal components to your regression model, you can account for more of the variability in the original data set. That is, your R^2 will increase. But in the same way that in backward elimination you start with a model that includes all the variables, thus having an R^2 of 100%, you need to consider the adjusted R^2. The question of how many principal components to include in the model is an important one, and there are many schools and methods to determine the correct amount.

One such method is parallel analysis. There is an R package called fa.parallel that can be used for this method. The idea is to simulate a data set similar to the original, another n x p matrix with different values, and then look at how much variability in the randomized matrix is explained by the principal components. It then plots this comparison in a scree plot. When the principal components start to explain less than or an equal amount of the variability for the original data

set compared to the random matrix, you disregard those later principal components. With the data from Green Square Capital, the parallel analysis deemed that 4 principal components were statistically significant and should be used in the regression model.

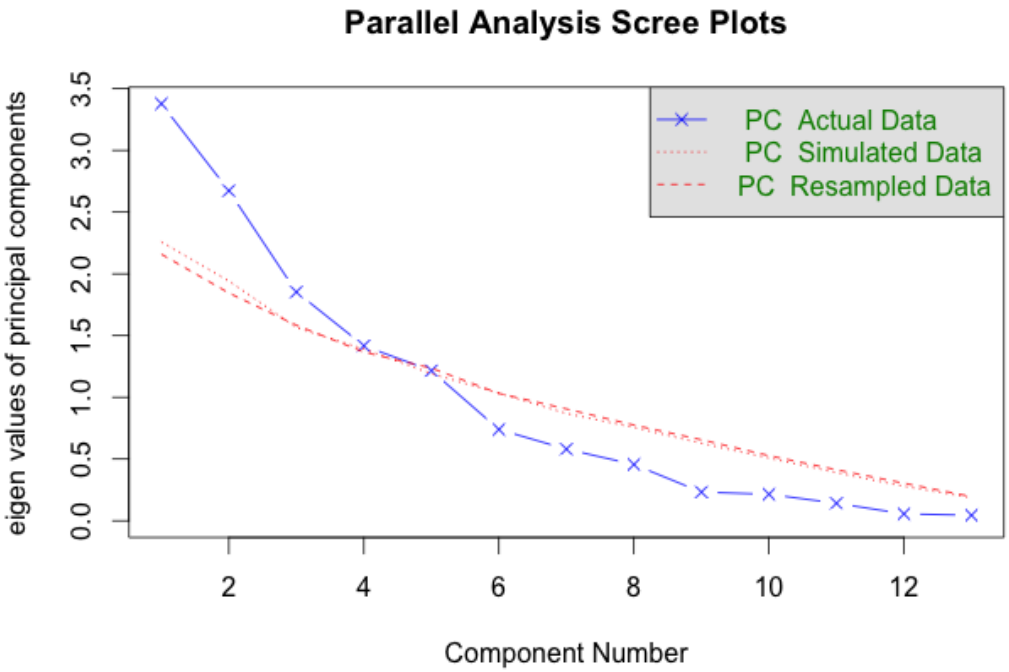


Figure 5: Scree Plot of Principal Components of Traded Companies

The regression model using those 4 principal components as predictors had an adjusted R^2 of about 50%. Since it only used 4 principal components it achieved the goal of creating a simpler model, but the adjusted R^2 was disappointing and much lower than what we were able to achieve with the model found using backward elimination. We used the same techniques to create a similar model for the dataset with Archer Daniels Midland (ADM) removed. In that instance, parallel analysis concluded that 5 principal components should be used in the model. This model has an adjusted R^2 of 74%. While this is considerably better and an arguably acceptable amount of variability to be explained, it is still lower than the 80% adjusted R^2 that we were able to achieve with our model from backward elimination.

As mentioned above, there are other methods in selecting how many principal components to include in the model. For example, you can select a desired R^2 (say 90%) and then continue to add principal components until you hit that target. Another option is to include principal components until they begin to level off. Since eigenvalues correspond to the percent of variability explained by each higher principal component, on the scree plot you can see where the eigenvalues are very close or level off. At this point, there are diminishing returns for adding more principal components.

Since we were disappointed by the adjusted R^2 achieved using parallel analysis, we did create other models using additional principal components. While we were able to increase our adjusted R^2 slightly (we were able to achieve 59% using 8 principal components for the original dataset, and 77% also with 8 principal components for the dataset excluding ADM), these models still had lower adjusted R^2 than the backward elimination models and included more predictors. We determined for this particular dataset and project, the models created using the backward elimination technique were superior.

```
Call:
lm(formula = ds$Prop.Change ~ yv1 + yv2 + yv3 + yv4)

Residuals:
     Min       1Q   Median       3Q      Max
-15.3666  -5.1046   0.6393   3.8595  25.9066

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.7026     1.3983   0.502 0.619285
yv1          -2.6341     0.7609  -3.462 0.001740 **
yv2           3.6051     0.8552   4.215 0.000235 ***
yv3          -0.6107     1.0273  -0.595 0.556935
yv4           2.7478     1.1747   2.339 0.026688 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 8.033 on 28 degrees of freedom
Multiple R-squared:  0.5596,    Adjusted R-squared:  0.4967
F-statistic: 8.895 on 4 and 28 DF,  p-value: 9.119e-05
```

Figure 6: Best Model on Traded Companies Using Principal Component Analysis

```
Call:
lm(formula = ds$Prop.Change ~ yv1 + yv2 + yv3 + yv4 + yv5)

Residuals:
     Min       1Q   Median       3Q      Max
-11.5143  -2.5672  -0.3381   3.1240   9.5818

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.1895     1.0078   0.188  0.85231
yv1          -2.6271     0.5452  -4.819 5.42e-05 ***
yv2           4.5379     0.6246   7.265 1.03e-07 ***
yv3           0.2931     0.7350   0.399  0.69330
yv4           2.5331     0.8329   3.041  0.00533 **
yv5          -2.2889     0.9132  -2.506  0.01879 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.701 on 26 degrees of freedom
Multiple R-squared:  0.7791,    Adjusted R-squared:  0.7366
F-statistic: 18.34 on 5 and 26 DF,  p-value: 8.565e-08
```

Figure 7: Best Model on Traded Companies with ADM Removed Using Principal Component Analysis

4. What's Next?

- We would like to run the data for more than one quarter to see whether our model does a good job in predicting percent change for future quarters. This would allow us to evaluate trends that may not be apparent in just one quarter.
- Consider other models that could capture more of the variability such as a multivariate times series model and evaluate 8 years worth of data.

References

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Variable Name	Code	Definition
Ticker	Ticker	NYSE Company Symbols
Sector	Sector	US Market Sectors
Price on 3/31/2016	Price 3/31	Equity Price - March 31,2016
Price on 6/30/2016	Price 6/30	Equity Price - June 30,2016
Proportion Change	Prop.Change	(Price 6/30 - Price 3/31)/(Price 6/30)
Market Cap	Mkt.Cap	Total Market Capitalization
Cash/Price	CP	Net Cash/Share Price
Debt/Equity	DE	Total Debt/Equity
AR/Revenue	ARR	Accounts Receivable/Revenue
Total Asset Turnover	TAT	Total Asset Turnover
DPS 5 Year Growth Rate	DPS	Dividend Per Share - 5 Year Growth Rate
Dividend Yield	DY	LTM Dividend Yield
EBITDA Margin	EBITDA.M	LTM EBITDA Margin
Unlevered FCF Margin	FCFM	Unlevered Free Cash Flow Margin
Price/Earnings	PE	Share to Price/LTM Earnings
Price/Book Value	PBV	Share Price/Book Value
Forward Price/Earnings	FPE	Share Price/Tangible Book Value
Diluted EPS - 5 year Growth Rate	DEPS	EPS-after all convertible securities executed
EPS Growth Rate	EPS	Earnings Per Share Growth Rate
Percent Change	Pct.Change	100 x Proportion Change

Table 1: Variables